

# Faderpunk WIP Apps — UX Cheatsheet

Synced to test/playground (IDs 29–43) · 13 Aug 2026 · A4 print

|                           |                                        |                                            |
|---------------------------|----------------------------------------|--------------------------------------------|
| <b>Main</b> — Fader alone | <b>Alt</b> — Shift + Fader             | <b>Third</b> — Channel button held + Fader |
| <b>Short</b> — Tap button | <b>Long</b> — Hold button (long-press) | <b>Shift+...</b> — Global Shift + gesture  |

**Jack CV Out / CV In** (Heat Pump, Grooves, Golden Gate, Arp de Lévy, Chord Vamp, Echolot, Loop de Cay; Super LFO is 2-ch with CV on Ch0):  
**CV Out** = app drives the jack · **CV In** = jack modulates a **CV Dest** (attenuated by **CV Att** 0–100% where present). Device-cycle Dest: Heat Pump & Super LFO via **Shift+Short** (Grooves: Configurator only). Golden Gate + Jack=CV In → MIDI-first (no CV/gate outs).

ID 29Heat Pumpready

Clock-synced sidechain ducking envelope

| FADERS                                                                                                                                                                                                                                                                                                                                                                                         | BUTTONS                                                                                                                                                                                                                                                                                                                               | LEDS / OTHER PARAMS                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li><b>Main</b> — Release (up = faster recovery)</li><li><b>Alt</b> — Duck depth (red LED)</li><li><b>Third</b> — Clock division (1/1 ... 1/32)</li></ul> <div>JACK PARAMS</div> <ul style="list-style-type: none"><li>Jack: CV Out / CV In</li><li>CV Dest: Depth · Release · Duck</li><li>Dest colors: Red · Yellow · Cyan</li><li>CV Att 0–100%</li></ul> | <ul style="list-style-type: none"><li><b>Short</b> — Manual duck trigger</li><li><b>Long</b> — Mute (freeze level; no snap to idle; dim red button)</li><li><b>Shift+Short (CV Out)</b> — Cycle clock division</li><li><b>Shift+Short (CV In)</b> — Cycle CV Dest (flash)</li><li><b>Shift+Long</b> — Invert duck direction</li></ul> | <ul style="list-style-type: none"><li>Button metronome Mid→Low per division · Mute = dim red (not off)</li><li>1/1 uses Color param</li><li>Invert: white ↔ off fade</li><li>Color, Range, MIDI CH/CC, NRPN, MIDI Out</li></ul> |

**Manual.** Clock-synced ducking envelope: each division pulse ducks, then recovers by Release. Depth sets how far. Mute freezes the envelope at its current level (clock triggers discarded). Invert flips duck direction. Third sets division directly; Shift+Short cycles division (CV Out) or CV Dest (CV In). CV Att scales modulation.

**FADERS**

- **Main** — Density (extra hits)
- **Alt** — Genre scrub (shared axis with Chord Vamp)
- **Third** — Swing (capped by Swing max %)

**GENRES**

- Dub → Disco → House → Techno → Trip-Hop → Hip-Hop → Jungle → UK Garage → Dubstep
- LEDs: continuous open red→yellow→green→cyan→blue spectrum (same as Chord Vamp)
- Sound morphs continuously between neighboring genres along the axis

**BUTTONS**

- **Short** — Mute (notes off)
- **Shift+Tap** — Genre fill until next downbeat (break at high density)
- **Shift+Hold** — Drum solo while held: 4-bar phrase (build → busier → peak → turnaround); resolves on release
- **Long** — Reset to downbeat

**JACK PARAMS**

- Jack: CV Out Any · CV Out Stacked · CV In Density · CV In Feel · CV In Reset
- Stacked sums families — hats 1, snare 2, kick 4 of 10 units; Open Hat counts as hats, Clap as snare, toms as kick
- The jack follows the groove, so it still pulses with every note switched off
- Range, CV Att

**PARAMS / LEDS**

- Notes: Kick, Snare, Hats, Open Hat, Low Tom, High Tom, Clap — note 0 switches a voice off
- MIDI Ch is the base for all seven; Ch Map overrides it per voice (7 packed nibbles, 0 = follow base)
- Groove, Swing max % (negative = reverse swing), GATE %, MIDI Out
- Groove = pocket: 0 is dead on the grid; up adds per-family ms drift (kick tight → hats loosest), velocity bow and walking ghosts
- GATE % is per voice — closed hats choke inside their step, open hats and toms may ring past it; CV pulse stays inside the 16th
- **Button Main**: genre hue sat half→full (density) + bar lightness; hit dips kick -50% / snare -30% / hats -10%
- **Button Third**: White→Color→Red (swing); same voice-weighted hit dips
- **Button Alt**: committed genre hue (Top/Bottom = live fader meters; color changes only after latch pickup)

**Manual.** Genre MIDI drums on the global clock (9 shared genres with Chord Vamp, including Jungle). Density reveals hits; Shift+Fader scrubs genre on the shared red→blue spectrum — button hue follows the latched genre (meters show live fader until pickup) while neighboring genres morph continuously; Button+Fader = Swing (capped by Swing max %). Shift+Tap plays a dedicated genre fill (or break at high density) that resolves on the next downbeat — Techno machine-guns, Hip-Hop and Jungle roll the snare, Dub and Dubstep answer sparsely — with velocity crescendoing toward the bar line, independent of the Density fader. Shift+Hold escalates into a drum solo that plays a four-bar phrase rather than looping the fill: an open build, a busier bar, a peak, then a turnaround that drops back to the sparse end-weighted figure and breathes. Each solo bar carries its own accent mask — the weight sits on the one and the backbeat, Feel only adds human jitter around it instead of randomising the dynamics — and every figure keeps rests and trades kick against snare rather than stacking them. Feel also opens ghost drags on the step right after a hit, so the fill-ins read as flams off the figure. Groove is the pocket control for the main pattern too, not only for the gestures: at zero every voice sits exactly on the grid, so Grooves locks tight against other Faderpunk apps. Turning it up opens a millisecond pocket in which each family drifts by its own amount — the kick stays tightest, the snare leans back into the backbeat, the hats carry the widest jitter — scaled by how tight the genre is, so Techno stays surgical while Hip-Hop and Trip-Hop drag audibly. Higher Groove also bows velocity across the bar and walks which ghost hits Density reveals, so repeated bars stop reading identically. Note lengths are per voice as well: closed hats choke inside their own step so they never smear, while open hats and toms are allowed to ring past the 16th; the CV pulse always stays inside the step even when the MIDI note is longer. Genre character stays intact: Techno rolls 16th kick bursts against clap stabs, Jungle chops the amen skeleton (kick on 1 and 3e, snare answering on the e), Dubstep keeps its half-time anchor, Dub and Trip-Hop stay wide open. Letting go drops straight into the end-weighted fill for the rest of the bar, so the solo always lands on the one. Releasing keeps either gesture running to the next downbeat so it always resolves. Above roughly 60 % density the same gesture reads subtractively and plays a break instead — the floor drops out (Techno keeps the kick, House the hats, Dub goes near-silent) and a late snare pickup hands the groove back on the one. Each genre carries three fill, three break, and three solo figures; low Feel locks to one, higher Feel rolls between them (same Die mix as Bassment). The button shows the gesture: bright white while filling, dim white while breaking. Watch the button: Main density ramps genre hue from half→full saturation with bar lightness; hits dim by voice (kick -50%, snare -30%, hats -10%) — not a clock blink. MIDI fires outside the clock subscriber (pending\_\* isolate). Third swing blends White→app→Red. Long resets to the downbeat. Four optional voices sit on top of the three written ones and cost no pattern data — each is derived from a hit the groove already plays, and each stays silent until you give it a note (note 0 = off, which is how they ship). Clap layers under the accented snare the way house and disco stack the two. Open Hat takes over an accented hat off the beat, muting the closed hat on that step rather than smearing both transients. The toms only appear inside a fill, break or solo, and only in the back half of the bar, where they lift the snare figure into the descending run that makes a fill read as a fill — High Tom through the third beat, Low Tom to land it. With all four off, Grooves plays exactly as it did before they existed. All seven voices share the base MIDI Ch unless Ch Map is non-zero; Ch Map packs one channel per voice into seven nibbles (voice order Kick, Snare, Hats, Open Hat, Low Tom, High Tom, Clap; nibble + 1 = channel), so drum machines that expect a channel per voice can be addressed without spending a param each. Swing max % carries the direction in its sign — negative swings the offbeats early. Jack folds the old Jack, Jack Mode and CV Dest into one enum, since a CV In destination and a CV Out level shape were never live at the same time. CV In modulates Density/Feel or triggers Reset; CV Att scales input.

**FADERS**

- **Main** — Fibonacci depth
- **Alt** — Cycle length N
- **Third** — Local speed grid (16th → quarter; center = follow Speed param)

**JACK PARAMS**

- Jack: CV Out / CV In (In = MIDI-first, no outs)
- CV Dest: Depth · Cycle · Reset
- Range, CV Att

**BUTTONS**

- **Short** — Reset to downbeat
- **Long** — Mute
- **Shift+Short** — Reverse gap direction
- **Shift+Long** — Cycle Mode (Note/CC/Pitch/Phi)

**MODE PARAM**

- Note · CC · Pitch (12-TET) · Phi ( $\phi$  pitch)
- Button color: default / orange / red / pink

**PARAMS**

- MIDI CH/Note/CC, GATE %, Speed, Color, MIDI Out

**Manual.** Fills cycle length N with Fibonacci gaps; deeper sets approach  $\phi$ . Mode param selects Note/CC/Pitch/Phi; Shift+Long cycles the same on-device. Third overrides Speed locally. Short resets; Shift+Short reverses travel. Jack=CV In drops CV/gate outs and uses Dest for Depth/Cycle/Reset.

**FADERS**

- **Ch0 Main** — Morph · **Alt** — Attenuation (Amp)
- **Ch0 Third** — Skew
- **Ch1 Main** — Symmetry · **Alt** — Speed
- **Ch1 Third** — Phase-Warp

**CV DESTINATIONS (CH0 SHIFT+SHORT)**

- Blue Rate Mod · Pink Phase · Red Reset · Cyan Amp
- Yellow Speed · Orange Morph · Violet Skew
- Green Warp · Rose Symmetry

**BUTTONS**

- **Ch0 Short** — Freeze toggle (white)
- **Ch0 Long** — In mute
- **Ch0 Shift+Short** — Cycle CV Dest
- **Ch0 Shift+Long** — Cycle Mix Mode
- **Ch1 Short** — Phase reset
- **Ch1 Long** — Out mute
- **Ch1 Shift+Short** — Phase reverse
- **Ch1 Shift+Long** — Clock sync toggle

**PARAMS**

- Speed, Range, MIDI, Grid Lock
- Mix Mode: Xfade / Min / Max / Sum
- Osc B: Quad / Octave · Mix balance 0–100%
- Rate-mod depth 0–100% (Configurator) · CV Dest can modulate it
- Ch0 jack = CV In · Ch1 jack = CV Out

**Manual.** Dual-osc morphing LFO. Ch0 Morph/Amp/Skew on CV In; Ch1 Symmetry/Speed/Warp on CV Out. Shift+Short on Ch0 cycles CV Dest. Freeze holds the wave; In-mute silences CV-in; Out-mute silences output. Ch1 Shift+Long toggles clock sync (Grid Lock). Mix balance 0–100% (Xfade only).

**FADERS**

- **Main** — Delay (up = shorter / faster)
- **Alt** — Feedback
- **Third** — Pitch interval (±24 st)

**BUTTONS**

- **Short** — Mute / ring-out
- **Long** — Panic (clear queue); skipped if Third used this hold

**LEDS**

- White flash on MIDI/gate in (~20% when muted)
- Ping-Pong: Top = Out A · Bottom = Out B

**PARAMS**

- I/O: MIDI→MIDI · MIDI→CV · CV→MIDI
- Delay: ms / Clock · Max delay
- Interval mode: Fixed / Stack / Bounce
- Routing: Single / Ping-Pong
- Signal: Pitch / Gate / CV→CC / Gate→Note
- MIDI In/Out A, Out Pong, CC, Note, Range, Color

**Manual.** Delay with feedback and optional pitch intervals. I/O picks MIDI/CV routing; Ping-Pong alternates MIDI Out A/B (MIDI→MIDI and CV→MIDI). Mute = ring-out (block new input; queue plays out). Long Panic clears the trail — Third interval edits never panic that hold. MIDI In always flashes white for diagnosis when muted.

**FADERS**

- **Main** — Evolve / mutation (0 = frozen)
- **Alt** — Texture (density / phrase / swing feel)
- **Third** — Lévy α / flight character

**JACK PARAMS**

- Jack: CV Out / CV In
- CV Out source (Jack=Out): Pitch · Gate · Velocity
- CV Dest (Jack=In): Evolve · Texture · Reroll
- Range, CV Att · V/Oct · Bypass quantizer

**BUTTONS**

- **Short** — Reroll note pool (+ reset)
- **Long** — Mute
- **Shift+Short (CV Out)** — Cycle CV Out source
- **Shift+Short (CV In)** — Reverse playback
- **Shift+Long** — Cycle octave span 1→2→3→4

**OCTAVE LED**

- Blue · Cyan · Yellow · Red

**LEDS**

- **Button / Top** — latch hue + level: Main = Color/Evolve · Alt = Orange/Texture · Third = Violet/α; mute = off
- **Bottom** — Pitch height in Color while gated (note activity)

**PARAMS**

- MIDI CH, Base Note, Color, MIDI Out
- Follow device tonic (default on)
- CV pitch + MIDI (dual path)

**Manual.** Generative arp whose pool evolves with Lévy-flight mutations. Main bottom freezes; raise to wander. Texture shapes phrase density/length; Third sets flight character (α). Shift+Long walks octave span (button color). Jack=CV Out: Shift+Short cycles Pitch/Gate/Velocity. Jack=CV In: Shift+Short reverses; CV modulates Evolve/Texture or fires Reroll. **Follow device tonic** (default on) makes the device Tonic a live transpose — hold the Scene button, move Fader 5, and the arp moves with Bassment, Contura, Chord Vamp and Venn. Because the pool holds absolute notes, it is *shifted* by the interval rather than rerolled, so the melodic shape it has evolved into survives the transpose. Takes effect at the next phrase start.

**FADERS**

- **Main (Perform)** — Scrub genre degrees × 3 octaves (pitch map)
- **Main (Auto)** — Tension (Meander only)
- **Alt** — Genre (+ Capture clip slot if clip exists)
- Shared axis: Dub → Disco → House → Techno → Trip-Hop → Hip-Hop → Jungle → UK Garage → Dubstep
- **Third (Auto)** — Feel (laid ↔ advanced stab length)
- **Third (Perform)** — Feel still stored; LEDs = blue / chord pitch

**JACK PARAMS**

- Jack: CV Out · CV In Macro · CV In Panic
- One enum — a CV In destination and CV Out were never live at once, and the freed slot is what lets Vamp grow
- Range, V/Oct, CV Att

**BUTTONS**

- **Perform hold** — Chord on; release = off (blue LEDs track pitch)
- **Auto short** — Play / pause
- **Shift+Short (Perform)** — Capture last N bars → Auto+Play
- **Shift+Short (Auto)** — Return to Perform
- **Auto Long** — Clear clip, reseed genre phrase
- **Shift+Long** — Panic (both modes)

**PARAMS**

- Root, Scale, Genre, Voicing, Velocity
- Follow device tonic (default on)
- Auto style: Repeat / Meander
- Start mode: Perform / Auto
- Capture: 16 / 8 / 4 / 2 / 1 bars
- LED: Perform = app color; Auto = orange; Hold Auto = rose (phrase pos)

**SCENE STORAGE**

- 8-slot vamp + Capture clip
- **Swing** 0–100 — genre-seeded; odd-16th delay (no Configurator param)
- Feel, scrub, tension, mode, play/pause

**Manual.** MIDI chord vamp with optional CV jack. Shared 9-genre axis with Grooves (Dub→...→Jungle→...→Dubstep). **Perform:** fader = genre-unique degrees ascending across ~3 octaves; hold plays the chord (blue = pitch). **Shift+Short** captures recent harmony (starts quantized to 16ths, lead-in cropped to first hit; hold lengths free) into a timed clip and arms Auto. **Auto:** genre rhythm phrase loops — multi-chord statement (~1–2 bars) then multi-bar rest; harmony Repeat or Meander; Feel sets short stab gate (rests fixed); Swing (scene, genre-seeded) delays odd 16ths on Auto + Capture. Short = play/pause; Long clears clip / reseeds; Shift+Short → Perform; Shift+Long = Panic in either mode. Alt = genre (+ Capture slot). CV Macro = scrub/tension; Panic from CV optional. **Follow device tonic** (default on) turns the device Tonic into a live transpose — hold the Scene button, move Fader 5, and the harmony moves with Bassment, Contura, Venn and Arp de Lévy; the pitch class comes from the device, the octave stays with Root, and it lands on the next chord. There is no scale-follow counterpart: Scale defines the progressions, and Vamp sits at the 16-param ceiling.

**FADERS**

- **Main** — Gate % (Note) or CC amount (CC mode)
- **Alt** — Clock division
- **Third** — Velocity / sensitivity

**BUTTONS**

- **Short** — Play / trigger held sample
- **Shift+Short** — Manual sample (capture now)
- **Long** (no fader move) — Soft mute (fade; no hard note-off)

**PARAMS / LEDS**

- Mode Note / CC, MIDI CH/CC, Color, Range, V/Oct
- Layer meters: IR→UV spectrum on Main/Alt/Third fader
- White flash on each sample; mute = red button

**Manual.** Clocked sample-and-hold. Capture on division or Shift+Short; Main shapes gate length or CC depth. Long soft-mutes so notes can ring out. Third sets how strongly velocity follows the sample.

**FADERS**

- **Main** — Chord type (Unison · Power · Min · Maj · Sus4 · Min7 · Dom7 · Maj7 · Min9 · Add9)
- **Alt** — Inversion (0–2) / open (top + octave)
- **Third** — Velocity

**BUTTONS**

- **Short** (no fader move) — Mute / unmute all voices
- **Alt+Short** — Cycle harmony octave (–1 / 0 / +1)
- **Alt+Long** — Step voicing preset

**PARAMS / LEDS**

- I/O: MIDI→MIDI · MIDI→CV · CV→MIDI; MIDI In/Out CH
- CV Gate: Sustain · ms · clock divisions
- Strum (ms), Color, Range, V/Oct, Bypass quantizer
- Near-0V CV = rest; unpatched stays silent until stable pitch
- LEDs: IR→UV spectrum per layer; mute = red; preset flash on Alt+Long

**Manual.** One root (last MIDI note or pitch CV) picks the chord root; the fader picks chord type (Unison · Power · Min · Maj · Sus4 · Min7 · Dom7 · Maj7 · Min9 · Add9); up to four MIDI voices sound (MIDI→CV follows the highest). Alt sets inversion (0–2) or open top. Not a polyphonic keyboard input. Mute is a short tap — use Third (button+fader) for velocity so mute and vel don't collide. Alt+Short cycles harmony octave; Alt+Long walks voicing presets (tight/dry → wide/slow). CV Gate: Sustain, fixed ms, or clock divisions. Near-0V CV = rest; unpatched float stays silent until a stable pitch.

**FADERS**

- **Main** — Pitch (span from Base Note); spectrum LED meter. In Pitch→MIDI it transposes the CV pitch, so centre = unchanged and an unpatched jack leaves the fader in charge
- **Alt** (Shift+Fader) — Decay strength: bottom = immortal, top ≈ 4 wraps (violet→red)
- **Third** (Btn+Fader; gate modes while muted) — Decay mode zones (IR→UV): Random Rose · Pitch Up Orange · Pitch Down Yellow · Vel+Root Lime · Pitch Root Cyan · Velocity Blue · Gate Violet

**BUTTONS**

- **Hold** (gate modes) — Note on; release = off (long notes OK). Hold+Fader = scale-degree glissando toward the fader (like Chord Vamp)
- **Shift+Short** — Cycle loop length 1→2→...→8 bars (amber→cyan flash)
- **Shift+Long** — Mute (red LED; decay pauses)
- **Short while muted** — Unmute (re-arms)
- **Shift+Long while muted** — Erase buffer (white flash)

**PARAMS / LEDS**

- Mode: Pitch→MIDI · Gate→MIDI · MIDI→MIDI · MIDI→CV · Fader→Both
- Decay Mode: Random · Pitch Up · Pitch Down · Vel+Root (default) · Pitch Root · Velocity · Gate
- MIDI CH / In / Out, Base Note, Span (semitones across fader)
- Color, Range, V/Oct
- Capture/playback hit: sharp button 80% dim (~2 PPQN) + Top white; layer meters on Top/Bottom/Button

**Manual.** Clocked overdub loop: play while armed to capture; each wrap applies Alt strength through the selected Decay Mode (0 = freeze forever). Pitch modes stay on the global scale. Random jitters scale pitch and only lowers velocity. Gate shortens duration toward a 16th floor. Hold the button for long notes — mute is Shift+Long. Hold+Fader glissandos through scale degrees toward the fader (Chord Vamp-style). Length is Shift+Short (1–8 bars), not Alt/Third. Third picks Decay Mode (Configurator sync). Quantized to 16ths.

**FADERS**

- **Main** — Attenuverter –100...+100 % (center deadband = 0)
- **Alt** (Shift+Fader) — Offset –100...+100 %
- **Third** (hold button + fader) — Slew 0...10 s

**BUTTONS**

- **Short click** (no fader move) — Mute toggle on release (sends Output Min)
- **Hold + fader** — edit Slew (suppresses mute)
- **Shift+Hold** — momentary Bypass (keeps Range / Min-Max / Rate)

**PARAMS / LEDS**

- Pipeline: Gate/CV → Att → Offset → Transfer → Slew → Steps → Min/Max → Rate
- Button LED only: Main = app color × out; Alt = red × |offset|; Third = cyan × slew (floor so hue stays visible at 0)
- Mute = off · Bypass = full · black blink on downbeat while out > Min

**Manual.** 1-channel CV→MIDI sculptor. Defaults are transparent (+100 % att, 0 offset, 0 slew, Linear, full CC window, 100 Hz). Gate mode runs a Schmitt threshold (0.1 V hysteresis) then the same pipeline. Mute sends Output Min once and goes silent; unmute resends the live value. Watch the button LED for level — fader LEDs stay dark. Uses clock ticker only (no CLOCK\_PUBSUB). No clock → no downbeat blink.

**FADERS**

- **Main** — F0 pulses A · F1 pulses B (density; absolute pulse count, Bjorklund redistributes)
- **Alt** (Shift) — F0 Extent (scale-degree span 0–24 from Note A) · F1 Interval (0–12 diatonic steps above the line for Ch2)
- **Third** (Btn1 or Btn2 hold) — F0 length A · F1 length B (2–32; independent)
- Rotation fixed at 0 (slots freed for Extent / Interval)
- **Division** (config) — note values 4/1 ... 1/32T (default 1/16)

**BUTTONS**

- **Btn1 or Btn2 hold** — Third (length A/B on F0/F1)
- **Btn2 tap** (no fader move) — Mute both outs
- **Shift+Btn1** — Cycle Logic OR → AND → XOR → Accnt
- **Shift+Btn2** — Toggle post-logic invert (shadow); white↔none fade like Heat Pump

**MELODY / LEDS**

- Note A (config) = degree 0 of the device quantizer key; Length-A walks an arch through 0...Extent scale degrees; Ch2 = same line + diatonic Interval
- Extent 0 = monotone Note A; higher = wider arch through the A cycle (device key)
- Division (note values), GATE %, Prob %, Humanize (Vel jitter; 0 = fixed max), MIDI CH / Note A / Interval B / Out, Color
- Color = App Color on Btn1; Btn0 = Logic hue; Top A/B = App Color / complement (layer hits)
- Main Btn: logic / app color · density bright · soft hit dip
- Alt: Top+Btn dim→sat ∝ Extent / Interval (fixed hues) · Third: shortest hue btn→complement ∝ length
- Top flashes = layer hits (Main) · Bottom flashes = logic outs (Main)
- Jack0 = logic result + melodic line · Jack1 coupled (OR→∩, AND→△, XOR→∩, Accnt→accent) + Interval

**Manual.** Two independent Euclidean masks combined with boolean logic — the OXI GEN dual-layer idea with a scale-aware melodic line. Color sets Layer A; Layer B LEDs use a fixed complement. Btn1 shows logic mode, not Color. Different lengths per layer enable polyrhythmic XOR/AND patterns Euclid cannot make alone. Melody: Note A is degree 0 of the device quantizer key. Extent is the arch span in scale degrees through Length-A; Interval B is diatonic steps above that line for Ch2 (wire indices 0–12 unchanged). Extent 0 = monotone Note A. Needs clock. Prob % < 100 adds optional flakiness; 100 is fully deterministic. No pitch CV (use GenSeq for that). Venn always draws its degrees from the device Scale. \*\*Follow device tonic\*\* (default on) adds the missing half: hold the Scene button and move Fader 5 and both lines transpose, together with Bassment, Contura, Chord Vamp and Arp de Lévy. The pitch class comes from the device, the octave stays with MIDI Note A, and the change lands at the start of the next A-cycle rather than mid-pattern.

**FADERS**

- **Main** — Density (extra onsets + freer degrees)
- **Alt** — Genre scrub (shared axis with Grooves / Chord Vamp)
- **Third** — Feel / Groove (swing · microtiming · velocity · ghosts · gate; capped by Groove max %)

**GENRES**

- Dub → Disco → House → Techno → Trip-Hop → Hip-Hop → Jungle → UK Garage → Dubstep
- Alt scrub LEDs: continuous IR→UV spectrum (shared palette)

**BUTTONS**

- **Short** — Mute (notes off)
- **Shift+Tap** — Bass fill into next bar's chord (pedal break at high Density); white button LED
- **Shift+Hold** — Bass solo (+1–2 oct), re-rolls each bar; resolves on the next downbeat
- **Long** — Next Voice (flash = voice color)
- Previous Voice: Configurator only

**JACK PARAMS**

- Jack: CV Out (pitch V/Oct) · CV In Density · CV In Feel · CV In Reset
- One enum — a CV In destination and CV Out were never live at once
- Range, V/Oct, CV Att
- Filter CC — cutoff number (0 = off; 74 is typical); brightness = genre base + Voice bite + accent dynamics, lifted by fill / solo
- Groove max % carries the swing direction in its sign — negative swings the offbeats early

**VOICES / LEDS**

- Order = historical era: bebop → Motown → P-Funk → fusion → dub duo → alt-funk → quirk
- **Mingus** Violet — chromatic neighbors
- **Jamerson** Orange — pocket root/fifth
- **Bootsy** Pink — funk ghosts + octave drops
- **Jaco** Cyan — melodic approaches + leaps
- **Robbie** Green — dub pocket, sustained / sparse
- **Flabba** Lime — roots walk, roomy gates
- **Flea** Red — staccato + aggressive jumps
- **Claypool** Yellow — quirky leaps + chromatic nudges
- Button: genre hue in play; Long flash = voice color; fill/solo = white (dim = pedal break); hits duck Mid→Low

**Manual.** Genre MIDI bass on the global clock (9 shared genres with Grooves / Chord Vamp — tropes live in `genre\_palette::GENRE\_PROG\_8`). Density reveals fill hits and freer scale degrees; Alt scrubs genre on the shared spectrum axis; Third is Feel / Groove (swing, microtiming, velocity contrast, ghost drag, gate character — depth capped by Groove max %). Notes sustain across empty 16ths until the next hit (legato when the pitch repeats). Phrases are 8 bars in AA B A' form: the rhythmic motif repeats, harmony walks the tropes, bars 5–6 answer (shifted, more approaches), bar 7 ornaments, bar 8 breathes (no ghosts/approaches). At higher Feel the line leaves "lock" mode: live Dice mix with the step-hash for approaches, ghosts, octave punches, and accent rake; pickups appear before downbeats, soft cores get dropped for air, ghosts can become dead notes, and approaches may be chromatic or scale-neighbor. Voice × genre soft-bias: pocket players stay sparse in Jungle/UKG; busy voices thin in Dub/Techno. \*\*Shift+Tap\*\* plays a dedicated bass fill that walks toward the "next" bar's chord (leading tone on the last hit) and resolves on the downbeat — Voice picks the contour (walk / arp / octave push / pedal); pocket Voices and Dub lean Pedal. At high Density the same gesture becomes a quiet pedal break (held root). \*\*Shift+Hold\*\* escalates into a register-lifted solo (+1 octave, +2 for Jaco/Claypool) with breath gaps and shorter gates; it re-rolls each bar while held and always resolves on the next downbeat. Button LED goes white for the gesture (dim = pedal break). Long cycles bassist Voice personas forward in historical era order (Mingus → Jamerson → Bootsy → Jaco → Robbie → Flabba → Flea → Claypool; button flash = voice color). Previous Voice lives in the Configurator; Swing Dir rides the sign of Groove max %. Short mutes. CV Out is pitch (V/Oct); CV In modulates Density / Feel or triggers Reset. \*\*Filter CC\*\* (0 = off; 74 is typical cutoff) tracks the line the same way the velocities do: a genre brightness lerp along the shared spectrum (Dub and Dubstep deep, UK Garage and Jungle cutting), the Voice persona's bite on top (Jamerson and Robbie stay warm, Flea and Claypool bright), and the hit's own dynamics so accents open the filter. Fills lift, solos bloom, the pedal break ducks away. MIDI note-ons and filter CC fire outside the clock subscriber (pending\_\* isolate), same pattern as Grooves. \*\*Follow device tonic\*\* (default on) makes the device Tonic a live transpose: hold the Scene button and move Fader 5, and Bassment moves with Contura, Chord Vamp, Venn and Arp de Lévy. It takes the pitch class from the device and keeps the octave of Root, so the line stays in its register; the new key lands on the next bar line. \*\*Follow device scale\*\* (default off) does the same for Scale — left off, the patch keeps its own, because Scale is part of the genre character.

**FADERS**

- **Main** — interval width (stepwise ↔ 2–3 octave leaps across the pool)
- **Alt** (Shift) — phrase length only (3–28 steps)
- **Third** (Btn hold) — note density (sparse rests ↔ continuous; audible mid)

**BUTTONS**

- **Short** — mute / unmute
- **Shift+Short** — octave span 1→2→3→4
- **Long** — next scale set
- **Shift+Long** — previous scale set

**SCALE SETS / LEDS**

- Flat list of named 12-TET sets (Ionian...Folk) — conventional interval-pattern labels, not cultural claims
- Follow device tonic (default on); Follow device scale uses the global Key mask instead of the selected set
- Per-set melodic feel (contour / leap / sustain) — expressivity is not a fader
- Button hue cycles by scale index · Top = gate · Bottom = pitch height · Division via config (default 1/8)
- Jack = CV Out (pitch) · CV In Density · CV In Interval · CV In Reset — one enum, since a CV In destination and CV Out were never live at once; CV Att scales input

**Manual.** Generates melodic contour phrases over selectable 12-TET pitch-class sets, anchored to the device quantizer tonic. Phrase starts and endings pull harder toward the tonic (cadence grammar); mid-phrase follows each set's ScaleFeel. Scale-set names are conventional labels for interval patterns (Western modes and other named collections on equal footing). They are not claims about living musical practice or authentic intonation. Needs a running clock. Jack = CV Out (pitch, 1V/oct) or CV In (Density / Interval / Reset + CV Att), set in the Configurator. Gates via MIDI note on/off. Phrases mix short, medium, and longer note lengths. \*\*Follow device tonic\*\* (default on) turns the device Tonic into a live transpose: hold the Scene button and move Fader 5 and Contura moves with Bassment, Chord Vamp, Venn and Arp de Lévy. The note pool is rebuilt on the change, so the cadence resolves in the new key; it takes effect at the next phrase start. \*\*Follow device scale\*\* (default off) swaps the Scale set for the device Scale.



**FADERS**

- **Main** — input trim on channel 0; attenuverter (CV) or threshold (Gate/Trigger) on channels 1–3
- **Alt** (Shift) — input offset on channel 0; output offset (CV) or hysteresis (Gate/Trigger) on channels 1–3
- **Third** (Btn hold) — input slew on channel 0; output slew (CV), minimum gate length (Gate) or pulse length (Trigger) on channels 1–3
- **Channel 0 while the internal LFO runs** — the three layers become morph, LFO speed and skew instead of trim, offset and slew

**BUTTONS**

- **Channel 0 short** — freeze the conditioned input value
- **Channel 0 Shift + short** — while the internal LFO runs: free-running or clock-synced (the button flashes when sync engages)
- **Channels 1–3 short** — mute/unmute that output (Shift not held); fader scrub while the button is held cancels the tap
- **Channels 1–3 Shift + long** — cycle that out through CV → Gate → Trigger; the configurator follows

**SIGNAL / LEDS**

- One CV input on channel 0, three outputs B/C/D on channels 1–3
- Five equidistant open-spectrum hues from blue to orange identify each channel; out LED hue is by channel, not by CV/Gate/Trigger mode
- Channel 0 is blue while CV in is active and one step warmer when the internal LFO runs; B, C and D take the next three steps ending at orange
- With nothing patched into the input, channel 0 becomes an internal morphing LFO that feeds B/C/D — *Source* forces CV In or Internal LFO instead of that auto fallback
- *LFO Speed* shifts the whole speed range to Slow or Slowest
- CV mode uses the configured Range; Gate and Trigger modes drive the jack as a digital GPO and ignore Range
- Defaults: B = CV, C = Gate, D = Trigger ( $\pm 5$  V on the CV path)
- Channel 0 LEDs dim with the active source amplitude in the channel hue
- CV outs dim with the shaped output level; Gate/Trigger meter the conditioned input while low and flash full when high
- Button LEDs use the same channel hues and breathe with their signal level: channel 0 stays bright when frozen; channels 1–3 turn off when muted

**Manual.** One CV in, three shaped outs. Each out is independently CV, Gate or Trigger: CV gets attenuverter / offset / slew, Gate and Trigger get threshold / hysteresis / pulse length. Shift + long press on an out cycles it through the three modes right on the device, and the change is stored as a parameter, so the configurator stays in sync. Channel 0 freezes the conditioned input; B/C/D mute. With nothing patched into the input, channel 0 turns into a morphing LFO and drives B/C/D on its own — the same wave engine as Super LFO, with morph, speed and skew on the three channel 0 fader layers and Shift + short press for clock sync. The *\*Source\** parameter overrides that fallback if you would rather force CV In or Internal LFO, and *\*LFO Speed\** moves the whole rate range down to Slow or Slowest. LEDs use five equidistant hues along the open spectrum from blue to orange — one per channel identity, not per CV/Gate/Trigger mode. Channel 0 is blue while CV in is active and one step warmer when the internal LFO runs; B, C and D take the next three steps ending at orange. All four channels dim with signal amplitude.